Local Continuum Yang–Mills Limit from Particle Trajectories

Abstract

This article gives a local continuum-limit theorem for a deterministic flat regulator sampled from the stationary mean-field color profile furnished by `01_convergence`. The particle system supplies the continuum gauge profile; the regulator supplies the exact flat lattice geometry. The main result is a localized Wilson-to-Yang–Mills limit on compact spacetime windows, proved by deterministic small-loop expansion and ordinary orientation-wise Riemann summation.

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1 Introduction

This article studies the local continuum limit of a deterministic flat $SU(3)$ regulator whose link variables are sampled from the stationary mean-field gauge profile induced by the particle system of `01_convergence`. The finite lattice and its gauge data are

$$\mathcal{L}^{(N)} = (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}), \qquad \Xi^{(N)} : \mathcal{N}^{(N)} \to \mathbb{R}^4, \qquad \mathcal{C}_{\square}^{(N)},$$

with link variables $U_3^{(N)}(e)$, plaquette holonomies $U_3^{(N)}(P)$, and discrete field strengths

$$F_P^{(N)} = (ig_3 A_P^{(N)})^{-1} \log U_3^{(N)}(P).$$

The continuum argument is organized in four steps:

1. construct the stationary mean-field color profile and its induced continuum connection;
2. sample that connection on a deterministic flat regulator;
3. identify the plaquette logarithms with the continuum curvature by a small-loop expansion;
4. pass the localized Wilson sums to continuum integrals by orientation-wise Riemann convergence.

2 Particle-system preliminaries

Definition 2.1 (Trajectory notation for bridge quantities)**.** For each family index N, and each observation time $t \in \{0, \dots, T\}$, define

$$Z^{(N)}(t) := (X_1^{(N)}(t), V_1^{(N)}(t), \dots, X_N^{(N)}(t), V_N^{(N)}(t)) \in (\mathbb{R}^8)^N,$$

and

$$\mu_t^N := \frac{1}{N} \sum_{i=1}^{N} \delta_{(X_i^{(N)}(t), V_i^{(N)}(t))}, \qquad \rho_t^N := \mathrm{Law}(Z^{(N)}(t)), \qquad \rho_{t,1}^N := \text{first marginal of } \rho_t^N.$$

Let $(\bar{X}_i, \bar{V}_i)$ be an exchangeable nonlinear mean-field family with common one-particle law f_t, and empirical average

$$\bar{\mu}_t^N := \frac{1}{N} \sum_{i=1}^{N} \delta_{(\bar{X}_i(t), \bar{V}_i(t))}.$$

Finite invariant and mean-field stationary laws are denoted

$$\pi_N \in \mathcal{P}((\mathbb{R}^8)^N), \quad \pi_{N,1} \in \mathcal{P}(\mathbb{R}^8), \quad \pi_\infty \in \mathcal{P}(\mathbb{R}^8).$$

Assumption 2.2 (Kinetic-core input translated from 01)**.** (i) **Dimension and confining core.** The position and velocity variables are in $d = 4$. The confining potential $U \in C^2(\mathbb{R}^4)$ is globally regular and strongly coercive:

$$\begin{aligned} \|\nabla U(x) - \nabla U(y)\| &\le L_U \|x - y\|, \\ \langle \nabla U(x) - \nabla U(y), x - y \rangle &\ge L_* \|x - y\|^2, \\ \langle x, \nabla U(x) \rangle &\ge m_U \|x\|^2 - \beta_U. \end{aligned}$$

(ii) **Regularized viscous kernel and companion rule.** For each N the interaction scale $\rho_N > 0$ defines

$$K_{\rho_N}(x, y) := \kappa_0 + \exp\left(-\frac{\|x - y\|^2}{2\rho_N^2}\right), \qquad K_{\rho_N}(x, y) \ge \kappa_0 > 0.$$

(iii) **Core drift and noise scales.** The finite-particle kinetic backbone has damping $\gamma > 0$, noise strength $\sigma > 0$, and viscous coupling $\nu_{\mathrm{visc}} \ge 0$, with Brownian forcing and dissipative drift as in the kinetic core of `01_convergence`.

(iv) **Velocity observation.** The velocity-squashing map is

$$\psi(v) := \frac{V_{\mathrm{alg}}}{V_{\mathrm{alg}} + \|v\|} v, \qquad V_{\mathrm{alg}} > 0.$$

Bridge assumptions

Assumption 2.3 (Particle-to-mean-field transfer and compactness)**.** Let $\{\mathcal{Q}^{(N)}\}_{N\geq N_0}$ be the sequence of trajectory systems from this paper, with empirical laws ρ_t^N on $(\mathbb{R}^4\times\mathbb{R}^4)^N$ and mean-field law f_t on $\mathbb{R}^4\times\mathbb{R}^4$. The following constants $L_U, L_*, \gamma, \sigma, \nu_{\mathrm{visc}}, C_T, C_{\mathrm{LSI}}, C_{\mathrm{tail}} > 0$ and random variables are fixed from the outset.

(i) **Confinement and kernel structure.** The confining potential and interaction kernel obey strong regularity and confining bounds:

$$U\in C^2,\qquad \|\nabla U(x)-\nabla U(y)\|\leq L_U\|x-y\|,\qquad \langle \nabla U(x)-\nabla U(y), x-y\rangle \geq L_*\|x-y\|^2,$$

and

$$K_\rho(x,y)=\kappa_0+\exp\left(-\frac{\|x-y\|^2}{2\rho^2}\right),\qquad K_\rho(x,y)\geq \kappa_0>0.$$

(ii) **Finite-time concentration / chaos.** For every finite horizon $T>0$ and every bounded-Lipschitz observable Φ, there is $C_T<\infty$ such that

$$\mathrm{KL}(\rho_t^N\|f_t^{\otimes N})\leq C_T,\qquad d_{\mathrm{TV}}(\rho_{t,1}^N, f_t)\leq \frac{C_T}{\sqrt{N}},\qquad 0\leq t\leq T,$$

and the first-particle Wasserstein transport estimate

$$\sup_{t\in[0,T]}\mathbb{E}\|X_1^{(N)}(t)-\bar{X}_1(t)\|^2+\mathbb{E}\|V_1^{(N)}(t)-\bar{V}_1(t)\|^2\leq \frac{C_T}{N}$$

holds for the finite/mean-field coupling.

(iii) **Finite-particle functional inequalities.** Each finite-N invariant law π_N satisfies a uniform hypocoercive logarithmic Sobolev inequality

$$\mathrm{Ent}_{\pi_N}(g^2)\leq C_{\mathrm{LSI}}\int \Gamma^N_{\mathrm{hypo}}(g,g)\,d\pi_N$$

for smooth cylinder functions g, yielding one-particle tv transport

$$d_{\mathrm{TV}}(\pi_{N,1},\pi_\infty)\to 0,\qquad \sup_N\int(\|x\|^2+\|v\|^2)\,\pi_{N,1}(dx\,dv)<\infty.$$

(iv) **Stationary compactness and wall-to-infinity tail control.** For every bounded continuous observable $\Phi:\mathbb{R}^8\to\mathbb{R}$ and every recentering sequence $a_R\in\mathbb{R}^4$ with $\|a_R\|\to\infty$,

$$\lim_{R\to\infty}\lim_{N\to\infty}\int\Phi(x-a_R,v)\,\pi_{N,1}^{a_R}(dx\,dv)=\int\Phi(x,v)\,\pi_\infty^0(dx\,dv),$$

where $\pi_{N,1}^{a_R}$ and $\pi_\infty^{a_R}$ are finite-N and mean-field stationary laws under translated confining envelopes.

Theorem 2.4 (Bridge package from 01_convergence)**.** *Assumption 2.2 yields all four items of Assumption 2.3, with explicit estimates as follows:*

(i) *Finite-time chaos and finite-time coupling: for every finite horizon $T > 0$ there is $C_T < \infty$ such that*

$$\mathrm{KL}(\rho_t^N \| f_t^{\otimes N}) \le C_T,$$
$$d_{\mathrm{TV}}(\rho_{t,1}^N, f_t) \le \frac{C_T}{\sqrt{N}},$$
$$\sup_{t\in[0,T]} \mathbb{E}\Big(\|X_1^{(N)}(t) - \bar{X}_1(t)\|^2 + \|V_1^{(N)}(t) - \bar{V}_1(t)\|^2 \Big) \le \frac{C_T}{N}.$$

(ii) *Finite-N hypocoercive regularization: there are $\lambda_* > 0$, $C_{\mathrm{LSI}} < \infty$ and $C_{\mathrm{TV}} < \infty$, all independent of N, such that*

$$\mathrm{Ent}_{\pi_N}(g^2) \le C_{\mathrm{LSI}} \int \Gamma_{\mathrm{hypo}}^N(g,g)\, d\pi_N, \qquad \mathrm{KL}(\mu P_t^N \| \pi_N) \le e^{-2\lambda_* t} \mathrm{KL}(\mu \| \pi_N),$$

there is also the TV contraction

$$d_{\mathrm{TV}}(\mu P_t^N, \pi_N) \le \sqrt{2}\, e^{-\lambda_* t} \mathrm{KL}(\mu \| \pi_N)^{1/2}$$

for all $\mu \ll \pi_N$, and by deterministic observation post-composition the same rate applies at observation times $t = m\tau$, where $\tau > 0$ is the fixed observation step used for the observed chain in `01_convergence` (equivalently one may write τ_N if a size-dependent label is preferred).

(iii) *Stationary compactness: there are constants $C_2 < \infty$, $N_0 < \infty$ such that*

$$\sup_{N \ge N_0} \int_{\mathbb{R}^8} (\|x\|^2 + \|v\|^2)\, \pi_{N,1}(dx\, dv) \le C_2,$$

and

$$\pi_{N,1} \Rightarrow \pi_\infty$$

in $\mathcal{P}(\mathbb{R}^8)$ as $N \to \infty$.

(iv) *Recentered limit and mean-field regularity: for every bounded $\Phi : \mathbb{R}^8 \to \mathbb{R}$ and every recentering sequence $a_R \in \mathbb{R}^4$, $\|a_R\| \to \infty$,*

$$\lim_{R\to\infty} \lim_{N\to\infty} \int_{\mathbb{R}^8} \Phi(x - a_R, v)\, \pi_{N,1}^{a_R}(dx\, dv) \to \int_{\mathbb{R}^8} \Phi(x,v)\, \pi_\infty^0(dx\, dv),$$

and π_∞ satisfies the mean-field inequality

$$\mathrm{Ent}_{\pi_\infty}(g^2) \le C_{\mathrm{LSI}} \int_{\mathbb{R}^8} \Gamma_{\mathrm{hypo}}^\infty(g,g)\, d\pi_\infty.$$

2.1 Imported labeled results from 01_convergence

The arguments below use several results from `01_convergence` by name. To keep the present paper self-contained at the theorem-reference level, we record here the imported statements under the labels used later in the file.

Theorem 2.5 (Imported finite de Finetti theorem on Polish path space)**.** *Let E be a Polish space and let $P^{(N)} \in \mathcal{P}(E^N)$ be exchangeable. Then for every $k \leq N$ there exists a mixing law*

$$Q_N \in \mathcal{P}(\mathcal{P}(E))$$

such that the k-particle marginal satisfies

$$d_{\mathrm{TV}}\left(P_k^{(N)},\ \int_{\mathcal{P}(E)} \eta^{\otimes k}\, Q_N(d\eta)\right) \leq \frac{k(k-1)}{2N}.$$

In `01_convergence` this is the Diaconis–Freedman finite de Finetti theorem applied on standard Borel state spaces.

Proposition 2.6 (Imported finite-dimensional quadratic moment bounds)**.** *Under Assumption 2.2, the N-particle kinetic system of `01_convergence` has uniform finite-dimensional moment bounds on every finite time horizon. In particular, for every $S > 0$ there exists $C_S < \infty$, independent of N, such that*

$$\sup_{0\leq t\leq S} \mathbb{E}\big[\|X_1^{(N)}(t)\|^2 + \|V_1^{(N)}(t)\|^2\big] \leq C_S,$$

and the corresponding fourth-moment increment bounds needed for Kolmogorov–Chentsov tightness hold on $C([0,S];\mathbb{R}^8)$.

Lemma 2.7 (Imported continuity of the mean-field force on moment classes)**.** *Let $F[\mu]$ denote the McKean–Vlasov viscous force field from `01_convergence`. On every class of probability measures on $\mathbb{R}^8$ with uniformly bounded second moments, the map*

$$\mu \longmapsto F[\mu]$$

is continuous in the topology used in the mean-field well-posedness argument; in particular it is stable under the weak convergence and moment bounds used in the path-space propagation-of-chaos proof below.

Proposition 2.8 (Imported McKean–Vlasov well-posedness)**.** *Under Assumption 2.2, the McKean–Vlasov equation of `01_convergence` is well posed on $\mathcal{P}_2(\mathbb{R}^8)$. In particular:*

(i) *for every initial law in $\mathcal{P}_2(\mathbb{R}^8)$ there exists a weak solution on every finite horizon;*

(ii) *that weak solution is unique in law.*

Proposition 2.9 (Imported stationary mean-field existence and uniqueness)**.** *Under Assumption 2.2, there exists a unique stationary McKean–Vlasov law*

$$\pi_\infty \in \mathcal{P}(\mathbb{R}^8)$$

with finite second moments. Writing ρ_∞ for its density and $\mathcal{L}_\mu$ for the mean-field kinetic generator

$$\mathcal{L}_\mu\phi(x,v) := v\cdot\nabla_x\phi(x,v) + \big(-\nabla U(x) - \gamma v + F[\mu](x,v)\big)\cdot\nabla_v\phi(x,v) + \frac{\sigma^2}{2}\Delta_v\phi(x,v),$$

the stationary law satisfies the weak stationary equation

$$\int_{\mathbb{R}^8} \mathcal{L}_{\pi_\infty}\phi(x,v)\,\pi_\infty(dx\,dv) = 0 \qquad \text{for every } \phi\in C_c^\infty(\mathbb{R}^8). \tag{1}$$

2.2 Thermodynamic transport of graph and plaquette observables

The next results transfer empirical trajectory information to the interaction graph and its elementary plaquettes.

Definition 2.10 (One-particle state and empirical stationary measure)**.** For each N, particle index i, and time slice $t \in \{0, 1, \ldots, T\}$, define the one-particle phase-space state

$$Y_i^{(N)}(t) := \big(X_i^{(N)}(t), V_i^{(N)}(t)\big) \in \mathbb{R}^8,$$

and the empirical same-time measure

$$L_t^{(N)} := \frac{1}{N}\sum_{i=1}^{N} \delta_{Y_i^{(N)}(t)}.$$

In the notation of Definition 2.1, $L_t^{(N)} = \mu_t^N$. For the path-space statements below, we use the same symbol

$$Y_i^{(N)}(s) \in \mathbb{R}^8, \qquad s \in [0, S],$$

for the continuous-time kinetic path of particle i under the N-particle dynamics of `01_convergence`, started from the invariant law π_N. The recorded discrete data are the observed values of this path at the integer-indexed sampling times.

Corollary 2.11 (Empirical-measure collapse at stationarity)**.** *Under Assumption 2.2, for every fixed recorded time $t \in \{0, 1, \ldots, T\}$, the empirical measure $L_t^{(N)}$ converges in probability to the unique stationary mean-field law π_∞:*

$$L_t^{(N)} \xrightarrow[N\to\infty]{\mathbb{P}} \pi_\infty \qquad \textit{in } \mathcal{P}(\mathbb{R}^8).$$

Proof. At $t = 0$, this is exactly the collapse of the de Finetti mixing law proved in the proof of the stationary mean-field limit theorem imported from `01_convergence`: with the canonical choice

$$Q_N := \mathrm{Law}\left(\frac{1}{N}\sum_{i=1}^{N} \delta_{Y_i^{(N)}(0)}\right),$$

the tightness–identification–uniqueness argument shows $Q_N \Rightarrow \delta_{\pi_\infty}$. Equivalently,

$$\frac{1}{N}\sum_{i=1}^{N} \delta_{Y_i^{(N)}(0)} \xrightarrow[N\to\infty]{\mathbb{P}} \pi_\infty.$$

Because the system is started from its invariant law π_N, the law of $L_t^{(N)}$ does not depend on t, so the same convergence holds for every fixed recorded slice $t \in \{0, 1, \ldots, T\}$. □

Definition 2.12 (Label-invariant pair rule and empirical edge measure)**.** Fix a recorded time slice $t \in \{0, 1, \ldots, T\}$. Define the same-time companion edge set by

$$\mathcal{P}_t^{(N)} := \{(i, j) \in \{1, \ldots, N\}^2 : i \neq j\}.$$

Assume that the same-time companion relation is generated by a label-invariant pair rule, in the sense that there exists a bounded measurable indicator

$$\chi_t : \mathbb{R}^8 \times \mathbb{R}^8 \to \{0, 1\}$$

such that for all N and all $i \neq j$,

$$\mathbf{1}_{\{(i,j)\in\mathcal{P}_t^{(N)}\}} = \chi_t\big(Y_i^{(N)}(t), Y_j^{(N)}(t)\big).$$

Define the empirical edge measure by

$$\Gamma_t^{(N)} := \frac{1}{N^2}\sum_{i\neq j} \mathbf{1}_{\{(i,j)\in\mathcal{P}_t^{(N)}\}}\,\delta_{\left(Y_i^{(N)}(t), Y_j^{(N)}(t)\right)}.$$

Lemma 2.13 (Weak convergence against bounded a.e.-continuous test functions)**.** *Let $\mu_N \Rightarrow \mu$ in $\mathcal{P}(E)$ and $f : E \to \mathbb{R}$ be bounded with discontinuity set $\operatorname{Disc}(f)$. If $\mu(\operatorname{Disc}(f)) = 0$, then*

$$\int_E f\, d\mu_N \to \int_E f\, d\mu.$$

This is the Portmanteau theorem for bounded functions with μ-a.e. continuity, see for example [1, Theorem 2.1] (the Portmanteau theorem).

Theorem 2.14 (Stationary thermodynamic limit of the interaction graph)**.** *Assume Definition 2.12. Let $\Phi : \mathbb{R}^8 \times \mathbb{R}^8 \to \mathbb{R}$ be bounded and continuous, and assume that $\chi_t\Phi$ is continuous at $(\pi_\infty \otimes \pi_\infty)$-almost every pair. Then*

$$\int_{\mathbb{R}^8\times\mathbb{R}^8} \Phi\, d\Gamma_t^{(N)} \xrightarrow[N\to\infty]{\mathbb{P}} \iint_{\mathbb{R}^8\times\mathbb{R}^8} \chi_t(z,z')\,\Phi(z,z')\,\pi_\infty(dz)\,\pi_\infty(dz').$$

Equivalently,

$$\frac{1}{N^2}\sum_{(i,j)\in\mathcal{P}_t^{(N)}} \Phi\big(Y_i^{(N)}(t), Y_j^{(N)}(t)\big) \xrightarrow[N\to\infty]{\mathbb{P}} \iint_{\mathbb{R}^8\times\mathbb{R}^8} \chi_t(z,z')\,\Phi(z,z')\,\pi_\infty(dz)\,\pi_\infty(dz').$$

If the number of recorded slices $T+1$ is fixed, the same convergence holds after summing over finitely many $t \in \{0, 1, \ldots, T\}$.

Proof. Write

$$\Psi_t(z,z') := \chi_t(z,z')\,\Phi(z,z').$$

By Definition 2.12,

$$\begin{aligned}
\int \Phi\, d\Gamma_t^{(N)} &= \frac{1}{N^2}\sum_{i\neq j}\Psi_t\big(Y_i^{(N)}(t), Y_j^{(N)}(t)\big)\\
&= \frac{1}{N^2}\sum_{i,j}\Psi_t\big(Y_i^{(N)}(t), Y_j^{(N)}(t)\big) - \frac{1}{N^2}\sum_{i=1}^{N}\Psi_t\big(Y_i^{(N)}(t), Y_i^{(N)}(t)\big)\\
&= \iint_{\mathbb{R}^8\times\mathbb{R}^8}\Psi_t(z,z')\,L_t^{(N)}(dz)\,L_t^{(N)}(dz') - R_t^{(N)},
\end{aligned}$$

where

$$|R_t^{(N)}| \le \frac{\|\Psi_t\|_\infty}{N}.$$

Hence $R_t^{(N)} \to 0$ in probability.

By Corollary 2.11,

$$L_t^{(N)} \xrightarrow{\mathbb{P}} \pi_\infty.$$

Since the limit is deterministic, the product measure $L_t^{(N)} \otimes L_t^{(N)}$ converges in probability to $\pi_\infty \otimes \pi_\infty$. Since $(\pi_\infty \otimes \pi_\infty)(\operatorname{Disc}\Psi_t) = 0$, where $\operatorname{Disc}\Psi_t$ is the discontinuity set, Lemma 2.13 yields

$$\iint \Psi_t(z,z')\, L_t^{(N)}(dz)\, L_t^{(N)}(dz') \xrightarrow[N\to\infty]{\mathbb{P}} \iint \Psi_t(z,z')\, \pi_\infty(dz)\, \pi_\infty(dz').$$

Combining the last two displays proves the first claim. The second is the same identity written in index notation. If $T+1$ is fixed, summing over finitely many times preserves convergence in probability. $\square$

Definition 2.15 (Adjacent-time empirical path and plaquette measures)**.** Fix $0 \le t < T$ and define

$$\widetilde{Y}_i^{(N)}(t) := \big(Y_i^{(N)}(t), Y_i^{(N)}(t+1)\big) \in \mathbb{R}^{12}, \qquad M_t^{(N)} := \frac{1}{N}\sum_{i=1}^N \delta_{\widetilde{Y}_i^{(N)}(t)}.$$

Define the persistent-plaquette indicator by

$$\chi_{\Box,t}\big((z_0,z_1),(w_0,w_1)\big) := \chi_t(z_0,w_0)\,\chi_{t+1}(z_1,w_1),$$

and the empirical plaquette measure by

$$\Gamma_{\Box,t}^{(N)} := \frac{1}{N^2}\sum_{i\neq j} \chi_{\Box,t}\big(\widetilde{Y}_i^{(N)}(t), \widetilde{Y}_j^{(N)}(t)\big)\, \delta_{\left(\widetilde{Y}_i^{(N)}(t), \widetilde{Y}_j^{(N)}(t)\right)}.$$

Definition 2.16 (Stationary mean-field path law)**.** For every finite horizon $S > 0$, let $\Pi^\infty_{[0,S]}$ be the law on $C([0,S];\mathbb{R}^8)$ of the unique McKean–Vlasov solution with initial law π_∞. Because π_∞ is stationary, $\Pi^\infty_{[0,S]}$ is stationary under time shifts. For each integer $0 \le t \le \lfloor S \rfloor - 1$, define the adjacent-time mean-field law by

$$\Lambda_t^\infty := (\mathrm{ev}_{t,t+1})_\# \Pi^\infty_{[0,S]}, \qquad \mathrm{ev}_{t,t+1}(\gamma) := \big(\gamma(t), \gamma(t+1)\big).$$

Definition 2.17 (Finite-horizon shift-stationary path laws)**.** For finite horizon $S > 0$, set

$$\mathcal{S}^{\mathrm{stat}}_{[0,S]} := \bigcap_{t=0}^{\lfloor S \rfloor - 1} \big\{\eta \in \mathcal{P}(C([0,S];\mathbb{R}^8)) : (\mathrm{ev}_{t,t+1})_\#\eta = (\mathrm{ev}_{0,1})_\#\eta\big\}.$$

This is the class of one-particle path laws whose unit-time adjacent marginals on the discrete times $0, \dots, \lfloor S \rfloor - 1$ are stationary.

Lemma 2.18 (Shift-stationary class is closed under weak limits)**.** *Let $E := C([0,S];\mathbb{R}^8)$. The set*

$$\mathcal{S}^{\mathrm{stat}}_{[0,S]} \subset \mathcal{P}(E)$$

is Borel (indeed, closed). If $\eta_n \Rightarrow \eta$ in $\mathcal{P}(E)$ and $\eta_n \in \mathcal{S}^{\mathrm{stat}}_{[0,S]}$ for all n, then $\eta \in \mathcal{S}^{\mathrm{stat}}_{[0,S]}$. Moreover, if $Q_n \Rightarrow Q$ in $\mathcal{P}(\mathcal{P}(E))$ and $Q_n(\mathcal{S}^{\mathrm{stat}}_{[0,S]}) = 1$ for all n, then $Q(\mathcal{S}^{\mathrm{stat}}_{[0,S]}) = 1$.

Proof. For each $t \in \{0, \dots, \lfloor S \rfloor - 1\}$, the map

$$\eta \mapsto (\mathrm{ev}_{t,t+1})_{\#}\eta$$

from $\mathcal{P}(E)$ to $\mathcal{P}(\mathbb{R}^8 \times \mathbb{R}^8)$ is continuous under weak convergence of $\mathcal{P}(E)$. Hence

$$\eta \mapsto ((\mathrm{ev}_{t,t+1})_{\#}\eta,\, (\mathrm{ev}_{0,1})_{\#}\eta)$$

is continuous, so each set

$$\{\eta : (\mathrm{ev}_{t,t+1})_{\#}\eta = (\mathrm{ev}_{0,1})_{\#}\eta\}$$

is closed as a preimage of the diagonal in $\mathcal{P}(\mathbb{R}^8 \times \mathbb{R}^8) \times \mathcal{P}(\mathbb{R}^8 \times \mathbb{R}^8)$. Finite intersection gives that $\mathcal{S}^{\mathrm{stat}}_{[0,S]}$ is closed. The final statement for Q_n is the Portmanteau theorem applied to this closed set. □

Theorem 2.19 (Stationary path-space propagation of chaos)**.** *Under Assumption 2.2, for every finite horizon $S > 0$, let $\widehat{P}^{(N)}_{[0,S]}$ be the law on $C([0,S]; (\mathbb{R}^8)^N)$ of the N-particle kinetic system started from its invariant law π_N, and define the empirical path measure*

$$\mathfrak{M}^{(N)}_{[0,S]} := \frac{1}{N}\sum_{i=1}^{N} \delta_{Y_i^{(N)}(\cdot)} \in \mathcal{P}\big(C([0,S]; \mathbb{R}^8)\big).$$

Then

$$\mathfrak{M}^{(N)}_{[0,S]} \xrightarrow[N\to\infty]{\mathbb{P}} \Pi^{\infty}_{[0,S]} \qquad \text{in } \mathcal{P}\big(C([0,S]; \mathbb{R}^8)\big).$$

Equivalently, for every bounded continuous path functional $\Psi : C([0,S]; \mathbb{R}^8) \to \mathbb{R}$,

$$\frac{1}{N}\sum_{i=1}^{N} \Psi\big(Y_i^{(N)}(\cdot)\big) \xrightarrow[N\to\infty]{\mathbb{P}} \int_{C([0,S];\mathbb{R}^8)} \Psi(\gamma)\, \Pi^{\infty}_{[0,S]}(d\gamma).$$

In particular, for each integer $0 \le t \le \lfloor S \rfloor - 1$,

$$M_t^{(N)} = (\mathrm{ev}_{t,t+1})_{\#}\mathfrak{M}^{(N)}_{[0,S]} \xrightarrow[N\to\infty]{\mathbb{P}} \Lambda_t^{\infty}.$$

Proof. Write $E := C([0,S]; \mathbb{R}^8)$. Because the invariant law π_N is exchangeable and the N-particle dynamics are permutation-equivariant, the stationary path law $\widehat{P}^{(N)}_{[0,S]}$ on E^N is exchangeable. Let

$$\mathfrak{M}^{(N)}_{[0,S]}(\omega) := \frac{1}{N}\sum_{i=1}^{N} \delta_{Y_i^{(N)}(\cdot)(\omega)} \in \mathcal{P}(E)$$

be the empirical measure of particle trajectories under $\widehat{P}^{(N)}_{[0,S]}$. Since E is Polish, hence standard Borel, Theorem 2.5 from `01_convergence`, i.e. the Diaconis–Freedman finite de Finetti theorem of [2], applies directly on the path space $E = C([0,S]; \mathbb{R}^8)$, yielding for every $k \le N$ a mixing law $\widehat{Q}_N \in \mathcal{P}(\mathcal{P}(E))$ such that

$$\widehat{Q}_N := \mathrm{Law}\big(\mathfrak{M}^{(N)}_{[0,S]}\big) \in \mathcal{P}(\mathcal{P}(E)),$$

and

$$d_{\mathrm{TV}}\left(\widehat{P}^{(N)}_{[0,S],k},\, \int_{\mathcal{P}(E)} \eta^{\otimes k}\, \widehat{Q}_N(d\eta)\right) \le \frac{k(k-1)}{2N}.$$

Hence the fixed-k path marginals of $\widehat{P}^{(N)}_{[0,S]}$ are within $O(1/N)$ in total variation of mixtures of product path laws. In particular, for $k=1$, $\widehat{P}^{(N)}_{[0,S],1} \in \mathcal{S}^{\mathrm{stat}}_{[0,S]}$, and Lemma 2.18 applies to weak limits of such stationary one-particle path laws. This records that adjacent-time stationarity is stable under the one-particle weak-limit passage. For the second-level laws $\widehat{Q}_N$, however, we do not use any support claim on $\mathcal{S}^{\mathrm{stat}}_{[0,S]}$; the identification of subsequential limits below proceeds instead through the nonlinear martingale problem and the uniqueness of the McKean–Vlasov path law for the initial distribution π_∞.

The one-particle path marginals are tight on E. Indeed, stationarity gives the same uniform moment bounds at every time as in the bridge package from `01_convergence` (in particular Assumption 2.3(ii) and (iii)), and Proposition 2.6 of `01_convergence` (the finite-dimensional quadratic moment bounds) provides the required uniform-in-N bounds. In particular, combining the linear-growth drift, additive velocity noise, and Burkholder–Davis–Gundy estimates for the velocity increments gives

$$\mathbb{E}\|Y_1^{(N)}(t) - Y_1^{(N)}(s)\|^4 \le C|t-s|^2,$$

uniformly in N, which implies tightness of $\mathrm{Law}(Y_1^{(N)}(\cdot))$ in E by the Kolmogorov–Chentsov criterion (see [1, Chapter 2]). Hence tightness of the second-level laws $\{\widehat{Q}_N\}_{N\ge 2}$. Since E is Polish, $\mathcal{P}(E)$ is Polish under the weak topology, and therefore $\{\widehat{Q}_N\}_{N\ge 2}$ is tight in $\mathcal{P}(E)$. Let $\widehat{Q}_\infty$ be any subsequential limit.

We identify $\widehat{Q}_\infty$-a.e. limit paths by the nonlinear martingale problem. Fix $\phi \in C_c^\infty(\mathbb{R}^8)$, times $0 \le u \le v \le S$, and a bounded continuous functional G depending only on the path segment on $[0,u]$. For $\mu \in \mathcal{P}(\mathbb{R}^8)$, write

$$\mathcal{L}_\mu \phi(x,v) := v \cdot \nabla_x \phi(x,v) + \big(-\nabla U(x) - \gamma v + F[\mu](x,v)\big) \cdot \nabla_v \phi(x,v) + \frac{\sigma^2}{2} \Delta_v \phi(x,v),$$

where $F[\mu]$ is the mean-field viscous field from `01_convergence`. For the finite-N system,

$$\begin{aligned}\phi(Y_i^{(N)}(v)) - \phi(Y_i^{(N)}(u)) - \int_u^v \Big[&\mathcal{L}_{\mathfrak{M}^{(N)}_{[0,S],s}} \phi\big(Y_i^{(N)}(s)\big) \\ &+ \big(F_i^{(N)}(s) - F[\mathfrak{M}^{(N)}_{[0,S],s}]\big(Y_i^{(N)}(s)\big)\big) \cdot \nabla_v \phi\big(Y_i^{(N)}(s)\big)\Big]\, ds\end{aligned}$$

is a martingale, where

$$\mathfrak{M}^{(N)}_{[0,S],s} := (\mathrm{ev}_s)_\# \mathfrak{M}^{(N)}_{[0,S]} = \frac{1}{N} \sum_{j=1}^N \delta_{Y_j^{(N)}(s)}.$$

Averaging over i and multiplying by $G(Y_i^{(N)}|_{[0,u]})$ gives zero expectation. The force-defect estimate from `01_convergence`, integrated on $[u,v]$, shows that the defect term contributes only $O(N^{-1/2})$ in expectation. Passing to the limit and using the continuity of $\mu \mapsto F[\mu]$ on uniformly moment-bounded classes (Lemma 2.7 in `01_convergence`, with the moment class provided by Assumption 2.3(ii)), together with continuity of $\eta \mapsto \eta_s = (\mathrm{ev}_s)_\# \eta$, $\widehat{Q}_\infty$-almost every path law η satisfies

$$\int_E G(\gamma|_{[0,u]}) \Big[\phi(\gamma(v)) - \phi(\gamma(u)) - \int_u^v \mathcal{L}_{\eta_s} \phi(\gamma(s))\, ds\Big] \eta(d\gamma) = 0,$$

where $\eta_s := (\mathrm{ev}_s)_\# \eta$. Thus $\widehat{Q}_\infty$-almost every η solves the McKean–Vlasov martingale problem on $[0,S]$.

Because each $\widehat{P}^{(N)}_{[0,S]}$ is stationary in time (as the path law induced by the invariant measure π_N), stationarity of time marginals implies, for every N,

$$(\mathrm{ev}_0)_\# \widehat{Q}_N \;=\; \mathrm{Law}\Big((\mathrm{ev}_0)_\# \mathfrak{M}^{(N)}_{[0,S]}\Big) \;=\; \mathrm{Law}(L^{(N)}_0).$$

By Corollary 2.11 with $t = 0$, $L^{(N)}_0 \xrightarrow{\mathbb{P}} \pi_\infty$, hence $(\mathrm{ev}_0)_\# \widehat{Q}_N \Rightarrow \delta_{\pi_\infty}$ in $\mathcal{P}(\mathbb{R}^8)$. Since $\mathrm{ev}_0 : E \to \mathbb{R}^8$ is continuous, the pushforward map $\eta \mapsto (\mathrm{ev}_0)_\# \eta$ is continuous on $\mathcal{P}(E)$ in the weak topology (continuous mapping theorem). This gives

$$(\mathrm{ev}_0)_\# \widehat{Q}_\infty = \delta_{\pi_\infty},$$

so $\widehat{Q}_\infty$-almost every η satisfies $\eta_0 = \pi_\infty$. Assumption 2.3(iii) and weak convergence $\pi_{N,1} \Rightarrow \pi_\infty$ imply $\pi_\infty \in \mathcal{P}_2(\mathbb{R}^8)$, so we may apply this well-posedness statement with initial law π_∞: By Proposition 2.8(ii) of `01_convergence`, the McKean–Vlasov equation from that framework is well posed for every initial law in $\mathcal{P}_2(\mathbb{R}^8)$, hence there is a unique path law with initial law π_∞; by Definition 2.16, that unique law is $\Pi^\infty_{[0,S]}$. Hence

$$\widehat{Q}_\infty = \delta_{\Pi^\infty_{[0,S]}}.$$

Every subsequence has the same limit, so

$$\mathrm{Law}\big(\mathfrak{M}^{(N)}_{[0,S]}\big) \Rightarrow \delta_{\Pi^\infty_{[0,S]}},$$

which is equivalent to the stated convergence in probability. The final claim follows from the continuous mapping theorem for the evaluation map $\mathrm{ev}_{t,t+1}$. As $\mathrm{ev}_{t,t+1} : E \to \mathbb{R}^8 \times \mathbb{R}^8$ is continuous, this gives the claimed pushforward limit. $\square$

Theorem 2.20 (Stationary thermodynamic limit of plaquette observables)**.** *Let $\Phi : (\mathbb{R}^{12})^2 \to \mathbb{R}$ be bounded and continuous, and assume that $\chi_{\Box,t}\Phi$ is continuous at $(\Lambda^\infty_t \otimes \Lambda^\infty_t)$-almost every pair. Then*

$$\int_{(\mathbb{R}^{12})^2} \Phi \, d\Gamma^{(N)}_{\Box,t} \xrightarrow[N\to\infty]{\mathbb{P}} \iint_{(\mathbb{R}^{12})^2} \chi_{\Box,t}(\zeta,\zeta')\,\Phi(\zeta,\zeta')\,\Lambda^\infty_t(d\zeta)\,\Lambda^\infty_t(d\zeta').$$

Proof. Set

$$\Psi_{\Box,t}(\zeta,\zeta') := \chi_{\Box,t}(\zeta,\zeta')\,\Phi(\zeta,\zeta').$$

By Definition 2.15,

$$\begin{aligned}\int \Phi \, d\Gamma^{(N)}_{\Box,t} &= \frac{1}{N^2}\sum_{i\neq j} \Psi_{\Box,t}\big(\widetilde{Y}^{(N)}_i(t), \widetilde{Y}^{(N)}_j(t)\big) \\ &= \iint \Psi_{\Box,t}(\zeta,\zeta')\,M^{(N)}_t(d\zeta)\,M^{(N)}_t(d\zeta') \;-\; R^{(N)}_{\Box,t},\end{aligned}$$

where

$$|R^{(N)}_{\Box,t}| \le \frac{\|\Psi_{\Box,t}\|_\infty}{N}.$$

Hence $R^{(N)}_{\Box,t} \to 0$ in probability. Because

$$M^{(N)}_t \xrightarrow[N\to\infty]{\mathbb{P}} \Lambda^\infty_t$$

by Theorem 2.19, the product measure $M_t^{(N)} \otimes M_t^{(N)}$ converges in probability to $\Lambda_t^\infty \otimes \Lambda_t^\infty$. Since $(\Lambda_t^\infty \otimes \Lambda_t^\infty)(\mathrm{Disc}\,\Psi_{\square,t}) = 0$, where $\mathrm{Disc}\,\Psi_{\square,t}$ is the discontinuity set, Lemma 2.13 yields

$$\iint \Psi_{\square,t}(\zeta,\zeta')\, M_t^{(N)}(d\zeta)\, M_t^{(N)}(d\zeta') \xrightarrow[N\to\infty]{\mathbb{P}} \iint \Psi_{\square,t}(\zeta,\zeta')\, \Lambda_t^\infty(d\zeta)\, \Lambda_t^\infty(d\zeta').$$

Combining the last two displays proves the theorem. □

3 Continuum gauge profile

Definition 3.1 (Continuum color field and induced gauge curvature)**.** Let

$$c : \mathbb{R}^4 \to \mathbb{C}^3$$

be a differentiable color field. Its induced continuum connection is the $\mathfrak{h}_3$-valued one-form

$$A_\mu[c](x) := \frac{1}{2}\sum_{a=1}^{8} \Re\big(c(x)^\dagger \lambda_a\, \partial_\mu c(x)\big)\lambda_a, \qquad 0 \le \mu \le 3,$$

and its induced continuum curvature is

$$F_{\mu\nu}[c] := \partial_\mu A_\nu[c] - \partial_\nu A_\mu[c] + ig_3[A_\mu[c], A_\nu[c]].$$

Remark 3.2 (Convention used in this paper)*.* The formulas in this paper are written in the Hermitian generator convention: $A_\mu[c]$ and $F_{\mu\nu}[c]$ take values in the Hermitian traceless space $\mathfrak{h}_3$, and the corresponding $SU(3)$ transports are written as exponentials $\exp(iH)$ with $H \in \mathfrak{h}_3$. Proposition 6.15 below translates these objects to the standard anti-Hermitian $\mathfrak{su}(3)$ convention used in the usual statement of pure $SU(3)$ Yang–Mills theory.

4 Deterministic regulator scales

The deterministic regulator is indexed by the lattice spacing $a_N \downarrow 0$ and the box size $R_N \uparrow \infty$. The continuum limit is taken at fixed compact spacetime windows, so a_N governs local resolution while R_N ensures that every fixed compact set is eventually contained in the regulator box.

Definition 4.1 (Deterministic regulator scales)**.** A *deterministic regulator scale family* is a pair of positive sequences

$$\big(a_N\big)_{N\ge N_0}, \qquad \big(R_N\big)_{N\ge N_0},$$

such that

$$a_N \downarrow 0, \qquad R_N \uparrow \infty, \qquad R_N^4 a_N \to 0.$$

Remark 4.2*.* The parameter a_N is the regulator spacing and R_N is the infrared box size. The geometric continuum limit is the regime $a_N \downarrow 0$ with $R_N \uparrow \infty$.

5 Deterministic flat QFT lattice

This section fixes the deterministic lattice regulator used in the continuum argument.

Definition 5.1 (Finite parameters)**.** Fix once and for all the positive parameters

$$d := 4, \qquad \kappa_0 > 0, \qquad \nu_{\mathrm{visc}} \geq 0, \qquad g_3 > 0, \qquad \epsilon_c > 0.$$

Fix also a surjective linear map

$$\Pi_{\mathrm{col}} : \mathbb{R}^4 \to \mathbb{R}^3.$$

For each N, the lattice spacing a_N and box radius R_N are those of Definition 4.1. The smooth positive interaction kernel appearing in the stationary mean-field color profile is

$$K(x,y) := \kappa_0 + \exp\left(-\frac{|x-y|^2}{2}\right), \qquad x, y \in \mathbb{R}^4.$$

Remark 5.2. The parameters in Definition 5.1 are the kernel and color parameters used in the finite lattice gauge construction.

Definition 5.3 (Deterministic flat regulator lattice)**.** Let $(a_N)_{N \geq N_0}$ and $(R_N)_{N \geq N_0}$ be the regulator scales of Definition 4.1, and assume in addition that

$$\frac{R_N}{a_N} \in \mathbb{N}.$$

Define the reflection-symmetric flat lattice box

$$\Lambda_N := a_N \mathbb{Z}^4 \cap [-R_N, R_N]^4.$$

The node set is

$$\mathcal{N}^{(N)} := \Lambda_N,$$

and the flat spacetime embedding is the identity map

$$\Xi^{(N)} : \mathcal{N}^{(N)} \to \mathbb{R}^4, \qquad \Xi^{(N)}(z) := z.$$

Definition 5.4 (Directed coordinate edges)**.** Let $e_0, \ldots, e_3$ denote the standard Euclidean basis of $\mathbb{R}^4$. For each $z \in \Lambda_N$ and each $\mu \in \{0, 1, 2, 3\}$ such that $z + a_N e_\mu \in \Lambda_N$, define the oriented edge

$$e^{(N)}_{z,\mu} := (z \to z + a_N e_\mu).$$

The full directed edge set is

$$\mathcal{E}^{(N)} := \{e^{(N)}_{z,\mu} : z, z + a_N e_\mu \in \Lambda_N,\ 0 \leq \mu \leq 3\},$$

and the finite QFT lattice is

$$\mathcal{L}^{(N)} := (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}).$$

Definition 5.5 (Coordinate plaquettes)**.** For $z \in \Lambda_N$ and $0 \leq \mu < \nu \leq 3$ such that the four corners

$$z, \qquad z + a_N e_\mu, \qquad z + a_N e_\nu, \qquad z + a_N(e_\mu + e_\nu)$$

all lie in Λ_N, define the elementary coordinate plaquette

$$P^{(N)}_{z,\mu\nu} := \big(z \to z + a_N e_\mu \to z + a_N(e_\mu + e_\nu) \to z + a_N e_\nu \to z\big).$$

The family of all such plaquettes is

$$\mathcal{C}^{(N)}_\square := \{P^{(N)}_{z,\mu\nu} : z, z + a_N e_\mu, z + a_N e_\nu, z + a_N(e_\mu + e_\nu) \in \Lambda_N,\ 0 \leq \mu < \nu \leq 3\}.$$

Definition 5.6 (Deterministic plaquette geometry)**.** For $P = P^{(N)}_{z,\mu\nu} \in \mathcal{C}^{(N)}_{\square}$, define the base point

$$x^{(N)}_P := z,$$

the two edge displacements

$$\tau^{(N)}_{P,\mu} := a_N e_\mu, \qquad \sigma^{(N)}_{P,\nu} := a_N e_\nu,$$

the oriented area bivector

$$\Sigma^{(N),\alpha\beta}_P := \tau^{(N),\alpha}_{P,\mu}\sigma^{(N),\beta}_{P,\nu} - \tau^{(N),\beta}_{P,\mu}\sigma^{(N),\alpha}_{P,\nu},$$

and the plaquette area

$$A^{(N)}_P := \Big(\frac{1}{2}\sum_{\alpha,\beta=0}^{3} \Sigma^{(N),\alpha\beta}_P \Sigma^{(N),\alpha\beta}_P\Big)^{1/2}.$$

Remark 5.7*.* Definitions 5.3–5.6 specify the deterministic flat regulator used in the main continuum-limit proof.

5.1 Flat-refinement geometric package

Proposition 5.8 (Flat geometry by construction)**.** *The deterministic lattice family of Definitions 5.3–5.6 has exact flat regulator geometry. More precisely:*

(i) *every edge is a coordinate edge of Euclidean length a_N;*

(ii) *every plaquette is a coordinate square of Euclidean area a_N^2;*

(iii) *distinct coordinate directions are orthogonal;*

(iv) *each lattice site is incident to uniformly boundedly many edges and plaquettes;*

(v) *every compact subset of $\mathbb{R}^4$ contains $O(a_N^{-4})$ plaquette base points.*

Proof. Items (i)–(iii) are immediate from the definitions of the coordinate edges and plaquettes in the Euclidean basis of $\mathbb{R}^4$. Item (iv) follows because each lattice site meets only finitely many coordinate edges and plaquettes, with a bound depending only on the ambient dimension 4. Item (v) follows from the fact that the lattice $a_N\mathbb{Z}^4$ has point density a_N^{-4}. □

Lemma 5.9 (Exact plaquette area in the deterministic regulator)**.** *For every N and every $P \in \mathcal{C}^{(N)}_{\square}$,*

$$A^{(N)}_P = a_N^2.$$

Proof. Let $P = P^{(N)}_{z,\mu\nu}$. Then

$$\tau^{(N)}_{P,\mu} = a_N e_\mu, \qquad \sigma^{(N)}_{P,\nu} = a_N e_\nu, \qquad \langle e_\mu, e_\nu\rangle = 0 \quad (\mu \neq \nu).$$

Hence

$$\big(A^{(N)}_P\big)^2 = \|\tau^{(N)}_{P,\mu}\|^2\|\sigma^{(N)}_{P,\nu}\|^2 - \langle\tau^{(N)}_{P,\mu}, \sigma^{(N)}_{P,\nu}\rangle^2 = a_N^4.$$

□

Lemma 5.10 (Local plaquette complexity in the deterministic regulator)**.** *For every compact $\mathcal{K} \subset \mathbb{R}^4$, there exists $C_{\mathcal{K}} < \infty$ such that*

$$\#\{P \in \mathcal{C}_\square^{(N)} : x_P^{(N)} \in \mathcal{K}\} \le C_{\mathcal{K}} a_N^{-4} \qquad \forall N \ge N_0.$$

Proof. The lattice $a_N\mathbb{Z}^4$ contributes at most $C_{\mathcal{K}} a_N^{-4}$ points to a fixed compact set $\mathcal{K}$, and at each lattice site there are at most $\binom{4}{2} = 6$ coordinate plaquette orientations. □

Corollary 5.11 (Deterministic flat-refinement consequences)**.** *For every compact $\mathcal{K} \subset \mathbb{R}^4$, there exists $C_{\mathcal{K}} < \infty$ such that for every $N \ge N_0$,*

$$A_P^{(N)} = a_N^2 \quad \forall P \in \mathcal{C}_\square^{(N)}, \qquad \#\{P \in \mathcal{C}_\square^{(N)} : x_P^{(N)} \in \mathcal{K}\} \le C_{\mathcal{K}} a_N^{-4}.$$

Definition 5.12 (Orientation-wise plaquette Riemann measures)**.** For each $0 \le \mu < \nu \le 3$ and each observable $\varphi \in C_c(\mathbb{R}^4)$, define

$$\mu_{\square,\mu\nu}^{(N)}(\varphi) := \sum_{\substack{P = P_{z,\mu\nu}^{(N)} \\ P \in \mathcal{C}_\square^{(N)}}} \big(A_P^{(N)}\big)^2 \varphi\big(x_P^{(N)}\big).$$

Remark 5.13. On the deterministic regulator, the plaquette measures are ordinary coordinate Riemann sums, resolved by orientation.

6 Stationary mean-field color profile and sampled lattice field

Definition 6.1 (Stationary mean-field color profile)**.** Let π_∞ be the unique stationary mean-field law from Proposition 2.9, and write $\rho_\infty(x, v)$ for its density on $\mathbb{R}^8 = \mathbb{R}_x^4 \times \mathbb{R}_v^4$. Define the stationary spatial marginal

$$\rho_{\infty,x}(x) := \int_{\mathbb{R}^4} \rho_\infty(x, v)\ \mathrm{d}v,$$

the stationary first velocity moment

$$j_\infty(x) := \int_{\mathbb{R}^4} v\, \rho_\infty(x, v)\ \mathrm{d}v \in \mathbb{R}^4,$$

the stationary kernel mass

$$m_\infty(x) := \iint_{\mathbb{R}^8} K(x, y)\, \rho_\infty(y, q)\ \mathrm{d}y\ \mathrm{d}q,$$

$$J_\infty(x) := \iint_{\mathbb{R}^8} K(x, y)\, q\, \rho_\infty(y, q)\ \mathrm{d}y\ \mathrm{d}q \in \mathbb{R}^4,$$

the stationary alignment field

$$A_\infty(x) := \frac{J_\infty(x)}{m_\infty(x)} \in \mathbb{R}^4,$$

$$a_\infty(x) := \Pi_{\mathrm{col}} A_\infty(x) \in \mathbb{R}^3,$$

and the stationary raw color profile

$$\tilde{\mathbf{c}}_\infty(x) := a_\infty(x) \in \mathbb{C}^3.$$

Let

$$\mathbf{c}_\infty(x) := \frac{\tilde{\mathbf{c}}_\infty(x)}{\sqrt{\tilde{\mathbf{c}}_\infty(x)^\dagger \tilde{\mathbf{c}}_\infty(x) + \epsilon_c^2}}.$$

Proposition 6.2 (Vanishing stationary mean velocity). *The stationary mean-field law satisfies*

$$\int_{\mathbb{R}^8} v\,\pi_\infty(dx\,dv) = 0.$$

Equivalently,

$$\int_{\mathbb{R}^4} j_\infty(x)\ \mathrm{d}x = 0.$$

Proposition 6.3 (Smoothness of the stationary mean-field color profile). *The stationary density ρ_∞ is C^∞ and strictly positive on $\mathbb{R}^8$. Consequently*

$$A_\infty \in C^\infty(\mathbb{R}^4;\mathbb{R}^4), \qquad \mathbf{c}_\infty \in C^\infty(\mathbb{R}^4;\mathbb{C}^3).$$

Proof of Proposition 6.2. Fix $j \in \{0,1,2,3\}$. Let $\eta \in C_c^\infty(\mathbb{R}^8)$ satisfy $\eta \equiv 1$ on the unit ball and $\eta \equiv 0$ outside the ball of radius 2, and set

$$\phi_{R,j}(x,v) := x_j\,\eta\left(\frac{|x|^2+|v|^2}{R^2}\right).$$

Then $\phi_{R,j} \in C_c^\infty(\mathbb{R}^8)$, so the stationary weak equation (1) gives

$$\int_{\mathbb{R}^8} \mathcal{L}_{\pi_\infty}\phi_{R,j}(x,v)\,\pi_\infty(dx\,dv) = 0.$$

Because $\partial_{x_k}\phi_{R,j} = \delta_{jk} + O((|x|^2+|v|^2)^{1/2}/R)$ and $\nabla_v\phi_{R,j} = O((|x|^2+|v|^2)^{1/2}/R)$, one has

$$\mathcal{L}_{\pi_\infty}\phi_{R,j}(x,v) = v_j + r_{R,j}(x,v),$$

where $r_{R,j}$ is supported on $\{R \le (|x|^2+|v|^2)^{1/2} \le 2R\}$ and satisfies

$$|r_{R,j}(x,v)| \le \frac{C}{R}\big(1+|x|^2+|v|^2\big)$$

with C independent of R. By Theorem 2.4(iii), π_∞ has finite second moments, hence

$$\int_{\mathbb{R}^8} |r_{R,j}|\,d\pi_\infty \to 0.$$

Passing to the limit $R \to \infty$ therefore gives

$$\int_{\mathbb{R}^8} v_j\,\pi_\infty(dx\,dv) = 0.$$

Since this holds for each j, the vector identity follows. The identity for j_∞ is Fubini's theorem. □

Proof of Proposition 6.3. Proposition 2.9 provides a stationary weak solution of the kinetic Fokker–Planck equation with smooth coefficients. The diffusion vector fields are the velocity derivatives $\partial_{v_0},\dots,\partial_{v_3}$, and their commutators with the transport field generate the spatial derivatives $\partial_{x_0},\dots,\partial_{x_3}$. Hörmander hypoellipticity therefore applies, so the stationary weak solution admits a smooth density ρ_∞. The associated hypoelliptic heat kernel is strictly positive, and invariance forces the stationary density to be strictly positive as well.

Since K is smooth and Proposition 2.9 provides the required stationary moment bounds, differentiation under the integral sign is legitimate in the displayed formulas for m_∞, J_∞, and A_∞. The strict positivity of the kernel floor gives $m_\infty(x) \ge \kappa_0 > 0$. Since Π_{col} is linear, $\tilde{\mathbf{c}}_\infty = \Pi_{\mathrm{col}}A_\infty$ is smooth as well. The final normalization with $\epsilon_c > 0$ preserves smoothness. □

Definition 6.4 (Stationary equilibrium color field on $\mathbb{R}^4$)**.** The stationary equilibrium color field is the profile of Definition 6.1 itself:

$$\bar{\tilde{\mathbf{c}}}_\infty(x) := \tilde{\mathbf{c}}_\infty(x), \qquad \bar{\mathbf{c}}_\infty(x) := \mathbf{c}_\infty(x), \qquad x \in \mathbb{R}^4.$$

Proposition 6.5 (Regularity of the induced continuum connection and curvature)**.** *Let*

$$A_\mu^\infty := A_\mu[\bar{\mathbf{c}}_\infty], \qquad F_{\mu\nu}^\infty := F_{\mu\nu}[\bar{\mathbf{c}}_\infty].$$

Then

$$A^\infty \in C^\infty(\mathbb{R}^4; \mathfrak{h}_3), \qquad F^\infty \in C^\infty(\mathbb{R}^4; \mathfrak{h}_3).$$

Proof. By Proposition 6.3, the field $\mathbf{c}_\infty$ is C^∞ on $\mathbb{R}^4$, and $\bar{\mathbf{c}}_\infty = \mathbf{c}_\infty$ by Definition 6.4. The formula in Definition 3.1 expresses each A_μ^∞ as a finite linear combination of products of $\bar{\mathbf{c}}_\infty$ and its first derivatives, so $A^\infty \in C^\infty(\mathbb{R}^4; \mathfrak{h}_3)$. The same definition expresses each $F_{\mu\nu}^\infty$ as a finite linear combination of first derivatives of A^∞ and commutators of its components, hence $F^\infty \in C^\infty(\mathbb{R}^4; \mathfrak{h}_3)$. □

Proposition 6.6 (Infrared and ultraviolet control of the equilibrium gauge profile)**.** *Let*

$$G(x) := \exp\left(-\frac{|x|^2}{2}\right), \qquad A_\mu^\infty := A_\mu[\bar{\mathbf{c}}_\infty], \qquad F_{\mu\nu}^\infty := F_{\mu\nu}[\bar{\mathbf{c}}_\infty].$$

Then:

(i) $A_\infty, \nabla A_\infty, \nabla^2 A_\infty \in L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$*;*

(ii) $\mathbf{c}_\infty, \nabla \mathbf{c}_\infty, \nabla^2 \mathbf{c}_\infty \in L^\infty(\mathbb{R}^4)$*, and* $\nabla \mathbf{c}_\infty, \nabla^2 \mathbf{c}_\infty \in L^1(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$*;*

(iii) $A^\infty, \nabla A^\infty, F^\infty \in L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$*;*

(iv) *the continuum Yang–Mills density*

$$e_\infty(x) := \sum_{0 \le \mu < \nu \le 3} \operatorname{tr}\big((F_{\mu\nu}^\infty(x))^2\big)$$

belongs to $W^{1,1}(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$*, and*

$$\lim_{R \to \infty} \int_{|x|>R} e_\infty(x)\ \mathrm{d}x = 0.$$

Proof. By Theorem 2.4(iii), π_∞ has finite second moments. Hence $\rho_{\infty,x} \in L^1(\mathbb{R}^4)$, $j_\infty \in L^1(\mathbb{R}^4; \mathbb{R}^4)$, and

$$\int_{\mathbb{R}^4} |y|^2 \rho_{\infty,x}(y)\ \mathrm{d}y \ + \ \int_{\mathbb{R}^4} |j_\infty(y)|\ \mathrm{d}y < \infty.$$

By Proposition 6.2, $\int_{\mathbb{R}^4} j_\infty(y)\ \mathrm{d}y = 0$. Therefore

$$m_\infty(x) = \kappa_0 + (G * \rho_{\infty,x})(x), \qquad J_\infty(x) = (G * j_\infty)(x).$$

For every multi-index β with $1 \le |\beta| \le 2$,

$$\partial^\beta m_\infty = (\partial^\beta G) * \rho_{\infty,x}, \qquad \partial^\beta J_\infty = (\partial^\beta G) * j_\infty.$$

Since each $\partial^\beta G$ is a Schwartz function, Young's inequality gives

$$\partial^\beta m_\infty, \partial^\beta J_\infty \in L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4),$$

and the Riemann–Lebesgue lemma gives

$$\partial^\beta m_\infty, \partial^\beta J_\infty \in C_0(\mathbb{R}^4).$$

Moreover $m_\infty(x) \geq \kappa_0 > 0$ everywhere. Quotient differentiation therefore yields

$$A_\infty = \frac{J_\infty}{m_\infty}, \qquad \nabla A_\infty, \nabla^2 A_\infty \in L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4) \cap C_0(\mathbb{R}^4),$$

and since $J_\infty \in C_0(\mathbb{R}^4)$, also $A_\infty \in C_0(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4)$.

Because $\tilde{\mathbf{c}}_\infty = \Pi_{\mathrm{col}} A_\infty$, the same bounds hold for $\tilde{\mathbf{c}}_\infty$. The map

$$u \mapsto \frac{u}{\sqrt{u^\dagger u + \epsilon_c^2}}$$

is a smooth map $\mathbb{C}^3 \to \mathbb{C}^3$ with bounded first and second derivatives, so the chain rule gives the asserted bounds for $\mathbf{c}_\infty$, $\nabla \mathbf{c}_\infty$, and $\nabla^2 \mathbf{c}_\infty$.

Definition 3.1 expresses each A_μ^∞ as a finite linear combination of products of $\bar{\mathbf{c}}_\infty$ and $\partial_\mu \bar{\mathbf{c}}_\infty$, and each derivative $\partial_\nu A_\mu^\infty$ as a finite linear combination of products of $\bar{\mathbf{c}}_\infty$, $\nabla \bar{\mathbf{c}}_\infty$, and $\nabla^2 \bar{\mathbf{c}}_\infty$. Since $\bar{\mathbf{c}}_\infty = \mathbf{c}_\infty$, the previous bounds imply

$$A^\infty, \nabla A^\infty \in L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4) \cap C_0(\mathbb{R}^4).$$

The curvature formula

$$F_{\mu\nu}^\infty = \partial_\mu A_\nu^\infty - \partial_\nu A_\mu^\infty + i g_3 [A_\mu^\infty, A_\nu^\infty]$$

then shows $F^\infty \in L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$.

Repeating the same convolution argument with higher derivatives of G gives that the first three spatial derivatives of A_∞ and $\mathbf{c}_\infty$ belong to $L^1(\mathbb{R}^4) \cap L^\infty(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$. Consequently $\nabla F^\infty \in L^1(\mathbb{R}^4)$. Finally, each summand $\operatorname{tr}((F_{\mu\nu}^\infty)^2)$ is continuous and bounded by $C\|F_{\mu\nu}^\infty\|^2$. Because $F_{\mu\nu}^\infty \in L^1 \cap L^\infty$, its square belongs to L^1, hence $e_\infty \in L^1(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$. Moreover

$$\nabla e_\infty = 2 \sum_{0 \leq \mu < \nu \leq 3} \operatorname{tr}\big(F_{\mu\nu}^\infty \nabla F_{\mu\nu}^\infty\big) \in L^1(\mathbb{R}^4),$$

so $e_\infty \in W^{1,1}(\mathbb{R}^4)$. The vanishing of the tails is the absolute continuity of the integral. $\square$

Definition 6.7 (Deterministically sampled site colors)**.** For each lattice site $z = (z^0, z^1, z^2, z^3) \in \mathcal{N}^{(N)}$, define

$$\tilde{\mathbf{c}}_z^{(N)} := \bar{\tilde{\mathbf{c}}}_\infty(z) \in \mathbb{C}^3,$$

$$\mathbf{c}_z^{(N)} := \bar{\mathbf{c}}_\infty(z) = \mathbf{c}_\infty(z) \in \mathbb{C}^3.$$

Equivalently, for each component $\alpha \in \{1, 2, 3\}$,

$$\tilde{c}_z^{(N),\alpha} := \tilde{c}_\infty^\alpha(z), \qquad c_z^{(N),\alpha} := c_\infty^\alpha(z).$$

Remark 6.8. The finite regulator no longer uses the random companion graph to define its geometry. The particle system enters through the stationary mean-field profile $\mathbf{c}_\infty$, and the deterministic flat lattice samples that profile as a regularizing probe.

Lemma 6.9 (Color normalization)**.** *For every site $x \in \mathcal{N}^{(N)}$,*

$$\big(\mathbf{c}_x^{(N)}\big)^\dagger \mathbf{c}_x^{(N)} = \frac{\big(\tilde{\mathbf{c}}_x^{(N)}\big)^\dagger \tilde{\mathbf{c}}_x^{(N)}}{\big(\tilde{\mathbf{c}}_x^{(N)}\big)^\dagger \tilde{\mathbf{c}}_x^{(N)} + \epsilon_c^2} \in [0, 1).$$

In particular, the normalized color vector is well-defined and obeys

$$\|\mathbf{c}_x^{(N)}\| < 1.$$

Proof. The denominator is strictly positive because $\epsilon_c > 0$. The displayed identity is immediate from the definition. Since $r/(r + \epsilon_c^2) \in [0, 1)$ for every $r \geq 0$, the claim follows. □

6.1 Link variables and parallel transport

Definition 6.10 (Gell-Mann basis and Hermitian generator space)**.** The Gell-Mann matrices are

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Each λ_a is Hermitian and traceless, and

$$\operatorname{tr}(\lambda_a \lambda_b) = 2\delta_{ab}.$$

We write

$$\mathfrak{h}_3 := \{H \in \mathbb{C}^{3\times 3} : H^\dagger = H, \ \operatorname{tr} H = 0\}$$

for the Hermitian traceless generator space. The Lie algebra $\mathfrak{su}(3)$ is identified with $i\mathfrak{h}_3$ in the usual physics convention.

Definition 6.11 (Direct connection sampling on lattice edges)**.** Let $\bar{\mathbf{c}}_\infty$ be the stationary spacetime color field of Definition 6.4, and write

$$A_\mu^\infty := A_\mu[\bar{\mathbf{c}}_\infty].$$

For the coordinate edge $e = e_{z,\mu}^{(N)} = (z \to z + a_N e_\mu)$, define its midpoint

$$m_e := z + \frac{a_N}{2} e_\mu,$$

the sampled Hermitian generator

$$\Theta_e^{(N)} := g_3 \, a_N \, A_\mu^\infty(m_e) \in \mathfrak{h}_3,$$

and the corresponding link matrix

$$U_3^{(N)}(e) := \exp\!\big(i\Theta_e^{(N)}\big) \in SU(3).$$

For the reversed orientation, set

$$U_3^{(N)}(e^{-1}) := \big(U_3^{(N)}(e)\big)^{-1}.$$

Lemma 6.12 (Link transport lies in $SU(3)$)**.** *For every oriented edge $e \in \mathcal{E}^{(N)}$, the matrix $U_3^{(N)}(e)$ belongs to $SU(3)$.*

Proof. By Definition 3.1, each $A_\mu^\infty(x)$ lies in $\mathfrak{h}_3$. Hence $\Theta_e^{(N)} \in \mathfrak{h}_3$, so $i\Theta_e^{(N)}$ is skew-Hermitian and traceless. Therefore

$$U_3^{(N)}(e) = \exp\bigl(i\Theta_e^{(N)}\bigr)$$

is unitary with determinant

$$\det U_3^{(N)}(e) = \exp\bigl(i \operatorname{tr} \Theta_e^{(N)}\bigr) = 1.$$

□

6.2 Plaquette field strength and Wilson action

Definition 6.13 (Gauge maps and transformed links)**.** A gauge map at level N is a function

$$\Omega^{(N)} : \mathcal{N}^{(N)} \to SU(3).$$

Its action on site colors and links is

$$\mathbf{c}_x^{(N),\Omega} := \Omega^{(N)}(x)\, \mathbf{c}_x^{(N)},$$

$$U_3^{(N),\Omega}(x \to y) := \Omega^{(N)}(y)\, U_3^{(N)}(x \to y)\, \bigl(\Omega^{(N)}(x)\bigr)^{-1}.$$

Definition 6.14 (Plaquette holonomy, discrete field strength, and Wilson action)**.** For the coordinate plaquette

$$P = P_{z,\mu\nu}^{(N)} = \bigl(z \to z + a_N e_\mu \to z + a_N(e_\mu + e_\nu) \to z + a_N e_\nu \to z\bigr),$$

define the plaquette holonomy

$$U_3^{(N)}(P) := U_3^{(N)}(e_{z,\mu}^{(N)}) U_3^{(N)}(e_{z+a_N e_\mu,\nu}^{(N)}) \bigl(U_3^{(N)}(e_{z+a_N e_\nu,\mu}^{(N)})\bigr)^{-1} \bigl(U_3^{(N)}(e_{z,\nu}^{(N)})\bigr)^{-1}.$$

Whenever $A_P^{(N)} > 0$ and $U_3^{(N)}(P)$ lies in the principal logarithm domain, define the discrete field strength

$$F_P^{(N)} := \frac{1}{i g_3 A_P^{(N)}} \log U_3^{(N)}(P) \in \mathfrak{h}_3.$$

The Wilson action is

$$S_{\mathrm{W}}^{(N)} := \frac{6}{g_3^2} \sum_{P \in \mathcal{C}_\square^{(N)}} \left(1 - \frac{1}{3} \Re \operatorname{tr} U_3^{(N)}(P)\right).$$

Proposition 6.15 (Bridge from the Hermitian convention to the standard $\mathfrak{su}(3)$ convention)**.** *The continuum and lattice objects of this paper are exactly the standard $SU(3)$ Yang–Mills objects after the usual identification*

$$\mathfrak{su}(3) = i\mathfrak{h}_3.$$

More precisely:

(i) *For every differentiable color field $c : \mathbb{R}^4 \to \mathbb{C}^3$, define the anti-Hermitian connection and curvature by*

$$\mathcal{A}_\mu[c] := iA_\mu[c] \in \mathfrak{su}(3), \qquad \mathcal{F}_{\mu\nu}[c] := iF_{\mu\nu}[c] \in \mathfrak{su}(3).$$

Then

$$\mathcal{F}_{\mu\nu}[c] = \partial_\mu \mathcal{A}_\nu[c] - \partial_\nu \mathcal{A}_\mu[c] + g_3[\mathcal{A}_\mu[c], \mathcal{A}_\nu[c]],$$

which is the standard anti-Hermitian $SU(3)$ curvature formula.

(ii) *For the stationary continuum field F^∞ of this paper, the local and global Yang–Mills densities agree with the standard anti-Hermitian normalization:*

$$\operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big) = -\operatorname{tr}\big((\mathcal{F}^\infty_{\mu\nu}(x))^2\big), \qquad \mathcal{F}^\infty_{\mu\nu} := iF^\infty_{\mu\nu},$$

so the continuum functional $S^{\rm cont}_{\rm YM}[F^\infty]$ is exactly the usual positive $SU(3)$ Yang–Mills action written in the anti-Hermitian convention.

(iii) *For every lattice edge and plaquette, if*

$$\mathcal{A}^{(N)}_e := i\Theta^{(N)}_e \in \mathfrak{su}(3), \qquad \mathcal{F}^{(N)}_P := iF^{(N)}_P \in \mathfrak{su}(3),$$

then

$$U^{(N)}_3(e) = \exp(\mathcal{A}^{(N)}_e), \qquad U^{(N)}_3(P) = \exp\big(g_3 A^{(N)}_P \mathcal{F}^{(N)}_P\big),$$

so the Wilson link and plaquette variables are unchanged by passing from the Hermitian convention to the standard anti-Hermitian $\mathfrak{su}(3)$ convention.

Proof. For (i), $\mathcal{A}_\mu[c] = iA_\mu[c]$ is skew-Hermitian and traceless because $A_\mu[c] \in \mathfrak{h}_3$; the same holds for $\mathcal{F}_{\mu\nu}[c] = iF_{\mu\nu}[c]$. Using

$$[\mathcal{A}_\mu[c], \mathcal{A}_\nu[c]] = [iA_\mu[c], iA_\nu[c]] = -[A_\mu[c], A_\nu[c]],$$

one computes

$$\begin{aligned}\partial_\mu \mathcal{A}_\nu[c] - \partial_\nu \mathcal{A}_\mu[c] + g_3[\mathcal{A}_\mu[c], \mathcal{A}_\nu[c]] &= i\partial_\mu A_\nu[c] - i\partial_\nu A_\mu[c] - g_3[A_\mu[c], A_\nu[c]] \\ &= i\Big(\partial_\mu A_\nu[c] - \partial_\nu A_\mu[c] + ig_3[A_\mu[c], A_\nu[c]]\Big) \\ &= iF_{\mu\nu}[c] = \mathcal{F}_{\mu\nu}[c].\end{aligned}$$

For (ii), $\mathcal{F}^\infty_{\mu\nu} = iF^\infty_{\mu\nu}$ gives

$$(\mathcal{F}^\infty_{\mu\nu})^2 = -(F^\infty_{\mu\nu})^2,$$

hence

$$-\operatorname{tr}\big((\mathcal{F}^\infty_{\mu\nu})^2\big) = \operatorname{tr}\big((F^\infty_{\mu\nu})^2\big).$$

Therefore the continuum functional defined later in Definitions 7.3 and 7.4 is exactly the usual positive Yang–Mills action in anti-Hermitian $\mathfrak{su}(3)$ variables.

For (iii), Definition 6.11 gives $U^{(N)}_3(e) = \exp(i\Theta^{(N)}_e)$, so with $\mathcal{A}^{(N)}_e := i\Theta^{(N)}_e$ we get

$$U^{(N)}_3(e) = \exp(\mathcal{A}^{(N)}_e).$$

Likewise Proposition 6.16 gives

$$U^{(N)}_3(P) = \exp\big(ig_3 A^{(N)}_P F^{(N)}_P\big) = \exp\big(g_3 A^{(N)}_P \mathcal{F}^{(N)}_P\big),$$

with $\mathcal{F}^{(N)}_P := iF^{(N)}_P \in \mathfrak{su}(3)$. Thus the group-valued Wilson transport variables are the same objects under either convention. $\square$

Proposition 6.16 (Exact plaquette exponential identity and Wilson expansion)**.** *Let $P \in \mathcal{C}_\square^{(N)}$ be such that $A_P^{(N)} > 0$ and $U_3^{(N)}(P)$ lies in the principal logarithm domain. Then*

$$U_3^{(N)}(P) = \exp\big(ig_3 A_P^{(N)} F_P^{(N)}\big).$$

If in addition

$$\|g_3 A_P^{(N)} F_P^{(N)}\|_{\mathrm{op}} \le \delta$$

for some fixed $\delta > 0$, then there exists a constant $C_\delta < \infty$, depending only on δ and the matrix norm convention, such that

$$1 - \frac{1}{3}\Re \operatorname{tr} U_3^{(N)}(P) = \frac{g_3^2 (A_P^{(N)})^2}{6} \operatorname{tr}\big((F_P^{(N)})^2\big) + R_P^{(N)},$$

with

$$|R_P^{(N)}| \le C_\delta\, g_3^3 (A_P^{(N)})^3 \|F_P^{(N)}\|_{\mathrm{op}}^3.$$

Proof. By Definition 6.14,

$$F_P^{(N)} = \frac{1}{ig_3 A_P^{(N)}} \log U_3^{(N)}(P),$$

so multiplication by $ig_3 A_P^{(N)}$ gives

$$\log U_3^{(N)}(P) = ig_3 A_P^{(N)} F_P^{(N)}.$$

Exponentiating both sides yields the exact identity

$$U_3^{(N)}(P) = \exp\big(ig_3 A_P^{(N)} F_P^{(N)}\big).$$

Set

$$X_P^{(N)} := ig_3 A_P^{(N)} F_P^{(N)}.$$

Because $F_P^{(N)} \in \mathfrak{h}_3$, the matrix $X_P^{(N)}$ is skew-Hermitian and traceless. The exponential series gives

$$e^{X_P^{(N)}} = I + X_P^{(N)} + \frac{1}{2}(X_P^{(N)})^2 + \mathcal{R}_P^{(N)},$$

where the third-order remainder satisfies the standard matrix bound

$$\|\mathcal{R}_P^{(N)}\|_{\mathrm{op}} \le C_\delta \|X_P^{(N)}\|_{\mathrm{op}}^3$$

whenever $\|X_P^{(N)}\|_{\mathrm{op}} \le \delta$. Taking traces and using $\operatorname{tr} X_P^{(N)} = 0$, we obtain

$$\operatorname{tr} U_3^{(N)}(P) = 3 + \frac{1}{2}\operatorname{tr}\big((X_P^{(N)})^2\big) + \operatorname{tr} \mathcal{R}_P^{(N)}.$$

Since $(ig_3 A_P^{(N)})^2 = -g_3^2 (A_P^{(N)})^2$,

$$\operatorname{tr}\big((X_P^{(N)})^2\big) = -g_3^2 (A_P^{(N)})^2 \operatorname{tr}\big((F_P^{(N)})^2\big).$$

Therefore

$$\Re \operatorname{tr} U_3^{(N)}(P) = 3 - \frac{g_3^2 (A_P^{(N)})^2}{2} \operatorname{tr}\big((F_P^{(N)})^2\big) + \Re \operatorname{tr} \mathcal{R}_P^{(N)}.$$

Subtracting from 1 after division by 3 gives

$$1-\frac{1}{3}\Re\operatorname{tr}U_3^{(N)}(P)=\frac{g_3^2(A_P^{(N)})^2}{6}\operatorname{tr}\big((F_P^{(N)})^2\big)-\frac{1}{3}\Re\operatorname{tr}\mathcal{R}_P^{(N)}.$$

It remains to estimate the last term. In fixed dimension, $|\operatorname{tr}Y|\le 3\|Y\|_{\mathrm{op}}$ for every 3×3 matrix Y, so

$$\left|-\frac{1}{3}\Re\operatorname{tr}\mathcal{R}_P^{(N)}\right|\le\|\mathcal{R}_P^{(N)}\|_{\mathrm{op}}\le C_\delta\|X_P^{(N)}\|_{\mathrm{op}}^3.$$

Since $\|X_P^{(N)}\|_{\mathrm{op}}=g_3A_P^{(N)}\|F_P^{(N)}\|_{\mathrm{op}}$, the claimed bound follows. □

Theorem 6.17 (Deterministic curvature identification)**.** *Let $\bar{\mathbf{c}}_\infty$ be the stationary spacetime color field of Definition 6.4, and set*

$$A_\mu^\infty:=A_\mu[\bar{\mathbf{c}}_\infty],\qquad F_{\mu\nu}^\infty:=F_{\mu\nu}[\bar{\mathbf{c}}_\infty].$$

Then there exist $N_*\in\mathbb{N}$ and $C_*<\infty$ such that, for all $N\ge N_*$ and every plaquette $P=P_{z,\mu\nu}^{(N)}\in\mathcal{C}_\square^{(N)}$, the principal logarithm is well-defined and

$$\Theta_P^{(N)}:=-i\,\log U_3^{(N)}(P),\qquad\left\|\Theta_P^{(N)}-g_3a_N^2F_{\mu\nu}^\infty(z)\right\|\le C_*a_N^3,$$

$$\left\|F_P^{(N)}-F_{\mu\nu}^\infty(z)\right\|\le C_*a_N,\qquad\left|\operatorname{tr}\big((F_P^{(N)})^2\big)-\operatorname{tr}\big((F_{\mu\nu}^\infty(z))^2\big)\right|\le C_*a_N.$$

Proof. By Propositions 6.5 and 6.6,

$$\sup_{x\in\mathbb{R}^4}\big(\|A^\infty(x)\|+\|\nabla A^\infty(x)\|+\|\nabla^2A^\infty(x)\|\big)<\infty.$$

Set

$$M:=\sup_{x\in\mathbb{R}^4}\big(\|A^\infty(x)\|+\|\nabla A^\infty(x)\|+\|\nabla^2A^\infty(x)\|\big).$$

For $P=P_{z,\mu\nu}^{(N)}$, set

$$\begin{aligned}X_1&:=ig_3a_NA_\mu^\infty\left(z+\frac{a_N}{2}e_\mu\right),\\X_2&:=ig_3a_NA_\nu^\infty\left(z+a_Ne_\mu+\frac{a_N}{2}e_\nu\right),\\X_3&:=-ig_3a_NA_\mu^\infty\left(z+a_Ne_\nu+\frac{a_N}{2}e_\mu\right),\\X_4&:=-ig_3a_NA_\nu^\infty\left(z+\frac{a_N}{2}e_\nu\right).\end{aligned}$$

Then

$$U_3^{(N)}(P)=e^{X_1}e^{X_2}e^{X_3}e^{X_4}.$$

Because A^∞ is globally bounded, one has $\|X_r\|\le g_3Ma_N$. For N large this is smaller than the principal-log radius, and the four-factor Baker–Campbell–Hausdorff formula gives

$$\log\big(e^{X_1}e^{X_2}e^{X_3}e^{X_4}\big)=\sum_{r=1}^4X_r+\frac{1}{2}\sum_{1\le r<s\le 4}[X_r,X_s]+O(a_N^3),$$

with a constant depending only on g_3 and M.

Taylor expansion of A_μ^∞ and A_ν^∞ at the base point z gives

$$X_1 + X_2 + X_3 + X_4 = ig_3 a_N^2 \big(\partial_\mu A_\nu^\infty - \partial_\nu A_\mu^\infty\big)(z) + O(a_N^3).$$

Using the zeroth-order approximations

$$X_1 = ig_3 a_N A_\mu^\infty(z) + O(a_N^2), \quad X_2 = ig_3 a_N A_\nu^\infty(z) + O(a_N^2),$$

$$X_3 = -ig_3 a_N A_\mu^\infty(z) + O(a_N^2), \quad X_4 = -ig_3 a_N A_\nu^\infty(z) + O(a_N^2),$$

one obtains

$$\frac{1}{2} \sum_{1\le r<s\le 4} [X_r, X_s] = -g_3^2 a_N^2 [A_\mu^\infty(z), A_\nu^\infty(z)] + O(a_N^3).$$

Therefore

$$\log U_3^{(N)}(P) = ig_3 a_N^2 \Big(\partial_\mu A_\nu^\infty - \partial_\nu A_\mu^\infty + ig_3[A_\mu^\infty, A_\nu^\infty]\Big)(z) + O(a_N^3),$$

that is,

$$\log U_3^{(N)}(P) = ig_3 a_N^2 F_{\mu\nu}^\infty(z) + O(a_N^3).$$

This gives the first estimate after multiplication by $-i$.

Since $A_P^{(N)} = a_N^2$ by Lemma 5.9,

$$F_P^{(N)} = \frac{1}{ig_3 a_N^2} \log U_3^{(N)}(P) = F_{\mu\nu}^\infty(z) + O(a_N).$$

The second estimate follows. The trace-square estimate follows from bilinearity:

$$\operatorname{tr}\!\big((F_P^{(N)})^2\big) - \operatorname{tr}\!\big((F_{\mu\nu}^\infty(z))^2\big) = \operatorname{tr}\Big(\big(F_P^{(N)} - F_{\mu\nu}^\infty(z)\big)\big(F_P^{(N)} + F_{\mu\nu}^\infty(z)\big)\Big),$$

and both factors are uniformly bounded globally for large N. □

6.3 Gauge invariance

Proposition 6.18 (Finite gauge invariance of plaquette traces)**.** *For every plaquette $P \in \mathcal{C}_\square^{(N)}$ and every gauge map $\Omega^{(N)}$,*

$$\operatorname{tr} U_3^{(N),\Omega}(P) = \operatorname{tr} U_3^{(N)}(P).$$

Consequently,

$$S_{\mathrm{W}}^{(N),\Omega} = S_{\mathrm{W}}^{(N)}.$$

Proof. Write the plaquette as the closed four-step loop

$$v_0 \to v_1 \to v_2 \to v_3 \to v_4 = v_0.$$

Under the gauge action from Definition 6.13,

$$U_3^{(N),\Omega}(P) = \prod_{k=1}^{4} \Omega^{(N)}(v_k)\, U_3^{(N)}(v_{k-1} \to v_k)\, \big(\Omega^{(N)}(v_{k-1})\big)^{-1}.$$

All consecutive inner factors telescope, leaving

$$U_3^{(N),\Omega}(P) = \Omega^{(N)}(v_0)\, U_3^{(N)}(P)\, \big(\Omega^{(N)}(v_0)\big)^{-1}.$$

Trace cyclicity gives the first claim, and summing over all plaquettes gives the Wilson-action invariance. □

Definition 6.19 (Compact-local big-O notation)**.** Let $\mathcal{K} \subset \mathbb{R}^4$ be compact and let $m \geq 0$. For a sequence of scalar, vector, or matrix-valued functions R_N defined on $\mathcal{K}$, we write

$$R_N = O_{\mathcal{K}}(a_N^m)$$

if there exists a constant $C_{\mathcal{K}} < \infty$, independent of N, such that

$$\sup_{x \in \mathcal{K}} |R_N(x)| \leq C_{\mathcal{K}} a_N^m$$

for all sufficiently large N. For quantities attached to plaquettes or edges with base point or midpoint in $\mathcal{K}$, the same notation means a uniform bound over those plaquettes or edges.

6.4 Orientation-wise Riemann convergence

Proposition 6.20 (Deterministic orientation-wise Riemann theorem)**.** *For every* $0 \leq \mu < \nu \leq 3$ *and every* $\varphi \in C_c(\mathbb{R}^4)$,

$$\mu^{(N)}_{\square,\mu\nu}(\varphi) \longrightarrow \int_{\mathbb{R}^4} \varphi(x)\ \mathrm{d}x.$$

Proof. Let $\varphi \in C_c(\mathbb{R}^4)$, and choose a compact box $\mathcal{K} = [-M, M]^4$ containing $\operatorname{supp}\varphi$. Since $R_N \uparrow \infty$, one has $\mathcal{K} \subset [-R_N, R_N]^4$ for all sufficiently large N. For such N, Definition 5.12 and Lemma 5.9 give

$$\mu^{(N)}_{\square,\mu\nu}(\varphi) = \sum_{z \in a_N\mathbb{Z}^4 \cap \mathcal{K}} a_N^4 \varphi(z),$$

because every $z \in a_N\mathbb{Z}^4 \cap \mathcal{K}$ contributes exactly one (μ, ν)-plaquette once a_N is small enough that all four corners of the plaquette remain in $[-R_N, R_N]^4$. The right-hand side is the standard four-dimensional Riemann sum for φ, hence converges to $\int_{\mathbb{R}^4} \varphi(x)\ \mathrm{d}x$. □

7 Local continuum limit

Definition 7.1 (Deterministic continuum curvature)**.** Let

$$F^{\infty}_{\mu\nu} := F_{\mu\nu}[\bar{\mathbf{c}}_\infty]$$

be the curvature induced by the stationary spacetime color field $\bar{\mathbf{c}}_\infty$. Write

$$F^{\infty}_{\mu\nu}(x) = \frac{1}{2}\sum_{a=1}^{8} F^{\infty,a}_{\mu\nu}(x)\lambda_a,$$

and define the local Yang–Mills density

$$e_\infty(x) := \sum_{0 \leq \mu < \nu \leq 3} \operatorname{tr}\big((F^{\infty}_{\mu\nu}(x))^2\big).$$

Theorem 7.2 (Local continuum limit of the deterministic Wilson functional)**.** *Let* $\chi \in C_c(\mathbb{R}^4)$*, and define the localized Wilson functional*

$$S^{(N)}_{\mathrm{W},\chi} := \frac{6}{g_3^2} \sum_{P \in \mathcal{C}^{(N)}_\square} \chi\big(x^{(N)}_P\big) \left(1 - \frac{1}{3}\Re \operatorname{tr} U^{(N)}_3(P)\right).$$

Then

$$S^{(N)}_{\mathrm{W},\chi} \longrightarrow \sum_{0 \leq \mu < \nu \leq 3} \int_{\mathbb{R}^4} \chi(x)\ \operatorname{tr}\big((F^{\infty}_{\mu\nu}(x))^2\big)\ \mathrm{d}x = \frac{1}{4}\int_{\mathbb{R}^4} \chi(x) \sum_{\mu,\nu=0}^{3}\sum_{a=1}^{8} F^{\infty,a}_{\mu\nu}(x) F^{\infty,a}_{\mu\nu}(x)\ \mathrm{d}x.$$

Proof. Let $\mathcal{K} := \operatorname{supp}\chi$. By Theorem 6.17, for every orientation $0 \le \mu < \nu \le 3$ and every plaquette $P = P^{(N)}_{z,\mu\nu}$ with $z \in \mathcal{K}$,

$$F_P^{(N)} = F^\infty_{\mu\nu}(z) + O_{\mathcal{K}}(a_N), \qquad \operatorname{tr}\big((F_P^{(N)})^2\big) = \operatorname{tr}\big((F^\infty_{\mu\nu}(z))^2\big) + O_{\mathcal{K}}(a_N).$$

Moreover, since $A_P^{(N)} = a_N^2$ and $F^\infty_{\mu\nu}$ is bounded on $\mathcal{K}$, the principal-log bound required by Proposition 6.16 holds uniformly on $\mathcal{K}$ for all sufficiently large N. Therefore,

$$1 - \frac{1}{3}\Re \operatorname{tr} U_3^{(N)}(P) = \frac{g_3^2 a_N^4}{6}\ \operatorname{tr}\big((F^\infty_{\mu\nu}(z))^2\big) + O_{\mathcal{K}}(a_N^5).$$

Fix $\mu < \nu$. Summing the last identity over all plaquettes $P = P^{(N)}_{z,\mu\nu}$ with $z \in \mathcal{K}$, multiplying by $\frac{6}{g_3^2}\chi(z)$, and using Corollary 5.11, we obtain

$$\begin{aligned} \frac{6}{g_3^2} \sum_{\substack{P = P^{(N)}_{z,\mu\nu} \\ P \in \mathcal{C}^{(N)}_\square}} \chi(z) \left(1 - \frac{1}{3}\Re \operatorname{tr} U_3^{(N)}(P)\right) \\ = \sum_{\substack{P = P^{(N)}_{z,\mu\nu} \\ P \in \mathcal{C}^{(N)}_\square}} \chi(z) a_N^4 \operatorname{tr}\big((F^\infty_{\mu\nu}(z))^2\big) + O(a_N), \end{aligned}$$

because the number of plaquettes with base point in $\mathcal{K}$ is $O_{\mathcal{K}}(a_N^{-4})$.

Applying Proposition 6.20 to the compactly supported function

$$\varphi_{\mu\nu}(x) := \chi(x) \operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big)$$

gives

$$\sum_{\substack{P = P^{(N)}_{z,\mu\nu} \\ P \in \mathcal{C}^{(N)}_\square}} \chi(z) a_N^4 \operatorname{tr}\big((F^\infty_{\mu\nu}(z))^2\big) \longrightarrow \int_{\mathbb{R}^4} \chi(x) \operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big)\ \mathrm{d}x.$$

Summing over the six unordered pairs $\mu < \nu$ yields

$$S^{(N)}_{\mathrm{W},\chi} \longrightarrow \sum_{0 \le \mu < \nu \le 3} \int_{\mathbb{R}^4} \chi(x) \operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big)\ \mathrm{d}x.$$

Finally, using

$$\operatorname{tr}\big((F^\infty_{\mu\nu})^2\big) = \frac{1}{2}\sum_{a=1}^{8} \big(F^{\infty,a}_{\mu\nu}\big)^2$$

and the antisymmetry $F^{\infty,a}_{\nu\mu} = -F^{\infty,a}_{\mu\nu}$, one has

$$\sum_{0 \le \mu < \nu \le 3} \operatorname{tr}\big((F^\infty_{\mu\nu})^2\big) = \frac{1}{4}\sum_{\mu,\nu=0}^{3}\sum_{a=1}^{8} F^{\infty,a}_{\mu\nu} F^{\infty,a}_{\mu\nu}.$$

This is exactly the stated limit. $\square$

Definition 7.3 (Localized continuum Yang–Mills functional)**.** For $\chi \in C_c(\mathbb{R}^4)$, define

$$S^{\text{cont}}_{\text{YM},\chi}[F^\infty] := \sum_{0\le\mu<\nu\le3} \int_{\mathbb{R}^4} \chi(x)\ \operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big)\ \mathrm{d}x.$$

Equivalently,

$$S^{\text{cont}}_{\text{YM},\chi}[F^\infty] = \frac{1}{4}\int_{\mathbb{R}^4} \chi(x) \sum_{\mu,\nu=0}^{3}\sum_{a=1}^{8} F^{\infty,a}_{\mu\nu}(x)F^{\infty,a}_{\mu\nu}(x)\ \mathrm{d}x.$$

Definition 7.4 (Global continuum Yang–Mills functional)**.** Define

$$S^{\text{cont}}_{\text{YM}}[F^\infty] := \sum_{0\le\mu<\nu\le3} \int_{\mathbb{R}^4} \operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big)\ \mathrm{d}x.$$

Equivalently,

$$S^{\text{cont}}_{\text{YM}}[F^\infty] = \frac{1}{4}\int_{\mathbb{R}^4} \sum_{\mu,\nu=0}^{3}\sum_{a=1}^{8} F^{\infty,a}_{\mu\nu}(x)F^{\infty,a}_{\mu\nu}(x)\ \mathrm{d}x.$$

Theorem 7.5 (Global cutoff-free continuum limit of the deterministic Wilson action)**.** *Let $S^{(N)}_{\text{W}}$ be the Wilson action of Definition 6.14. Then*

$$S^{(N)}_{\text{W}} \longrightarrow S^{\text{cont}}_{\text{YM}}[F^\infty].$$

Proof. For $0 \le \mu < \nu \le 3$, write

$$e_{\mu\nu}(x) := \operatorname{tr}\big((F^\infty_{\mu\nu}(x))^2\big).$$

By Proposition 6.6, each $e_{\mu\nu}$ belongs to $W^{1,1}(\mathbb{R}^4) \cap C_0(\mathbb{R}^4)$.

By Theorem 6.17, there exist $N_* \in \mathbb{N}$ and $C_* < \infty$ such that, for every $N \ge N_*$ and every plaquette $P = P^{(N)}_{z,\mu\nu}$,

$$\|F^{(N)}_P\| \le \sup_{x\in\mathbb{R}^4} \|F^\infty(x)\| + C_* a_N.$$

Hence Proposition 6.16 applies uniformly for all sufficiently large N, and with $A^{(N)}_P = a_N^2$ gives

$$\left|\frac{6}{g_3^2}\left(1 - \frac{1}{3}\Re \operatorname{tr} U^{(N)}_3(P)\right) - a_N^4 e_{\mu\nu}(z)\right| \le C a_N^5,$$

for a constant $C < \infty$ independent of N, z, μ, and ν.

Summing over all (μ,ν)-plaquettes yields

$$\frac{6}{g_3^2}\sum_{\substack{P=P^{(N)}_{z,\mu\nu}\\ P\in\mathcal{C}^{(N)}_\square}} \left(1 - \frac{1}{3}\Re \operatorname{tr} U^{(N)}_3(P)\right) = \sum_{z\in a_N\mathbb{Z}^4\cap[-R_N,R_N]^4} a_N^4 e_{\mu\nu}(z) + O\big(a_N^5 \#(a_N\mathbb{Z}^4\cap[-R_N,R_N]^4)\big).$$

Since $\#(a_N\mathbb{Z}^4\cap[-R_N,R_N]^4) \le C' R_N^4 a_N^{-4}$, the scale condition $R_N^4 a_N \to 0$ from Definition 4.1 implies that the remainder is $o(1)$.

It remains to pass the global Riemann sums to the integral. Partition $\mathbb{R}^4$ into the half-open cubes $Q_{N,z} := z + [0,a_N)^4$. For each $z \in a_N\mathbb{Z}^4$, the Sobolev fundamental theorem on $Q_{N,z}$ gives

$$\int_{Q_{N,z}} |e_{\mu\nu}(x) - e_{\mu\nu}(z)|\ \mathrm{d}x \le C a_N \int_{Q_{N,z}} |\nabla e_{\mu\nu}(x)|\ \mathrm{d}x.$$

Summing over all cubes indexed by plaquette bases z with $P^{(N)}_{z,\mu\nu} \in \mathcal{C}^{(N)}_{\square}$ (equivalently, those $z \in a_N\mathbb{Z}^4 \cap [-R_N, R_N]^4$ for which all four corners of the (μ,ν)-plaquette are in $[-R_N, R_N]^4$), and noting that discarded cubes only occupy a coordinate boundary slab of total width a_N gives only an $o(1)$ error.

$$\left| \sum_{z\in a_N\mathbb{Z}^4\cap[-R_N,R_N]^4} a_N^4 e_{\mu\nu}(z) - \int_{[-R_N,R_N]^4} e_{\mu\nu}(x)\,\mathrm{d}x \right| \le C a_N \|\nabla e_{\mu\nu}\|_{L^1(\mathbb{R}^4)}.$$

Since $e_{\mu\nu} \in L^1(\mathbb{R}^4)$ and $R_N \uparrow \infty$,

$$\int_{[-R_N,R_N]^4} e_{\mu\nu}(x)\,\mathrm{d}x \longrightarrow \int_{\mathbb{R}^4} e_{\mu\nu}(x)\,\mathrm{d}x.$$

Therefore

$$\frac{6}{g_3^2} \sum_{\substack{P=P^{(N)}_{z,\mu\nu}\\ P\in\mathcal{C}^{(N)}_{\square}}} \left(1 - \frac{1}{3}\Re\operatorname{tr} U_3^{(N)}(P)\right) \longrightarrow \int_{\mathbb{R}^4} e_{\mu\nu}(x)\,\mathrm{d}x.$$

Summing over the six unordered pairs $\mu < \nu$ gives

$$S_{\mathrm{W}}^{(N)} \longrightarrow \sum_{0\le\mu<\nu\le3} \int_{\mathbb{R}^4} \operatorname{tr}\big((F^{\infty}_{\mu\nu}(x))^2\big)\,\mathrm{d}x = S^{\mathrm{cont}}_{\mathrm{YM}}[F^{\infty}].$$

□

8 Exported theorem package for downstream papers

Theorem 8.1 (Export package of Paper 2)**.** *The results of the present paper exported to later papers in the series are the following.*

(i) ***Deterministic flat regulator package.*** *Definitions 4.1, 5.1, 5.3, and 5.6 construct the deterministic flat $SU(3)$ regulator family*

$$\mathcal{L}^{(N)} = (\mathcal{N}^{(N)}, \mathcal{E}^{(N)}, \mathcal{C}^{(N)}_{\square})$$

used in the continuum-limit argument.

(ii) ***Sampled lattice gauge sector.*** *Definitions 6.11 and 6.14, together with Proposition 6.15, construct the link variables $U_3^{(N)}(e) \in SU(3)$, plaquette holonomies $U_3^{(N)}(P) \in SU(3)$, and discrete field strengths $F_P^{(N)} \in \mathfrak{h}_3$ / $\mathcal{F}_P^{(N)} \in \mathfrak{su}(3)$.*

(iii) ***Smooth continuum gauge field.*** *Definition 3.1, Proposition 6.3, Proposition 6.5, and Proposition 6.15 construct the stationary continuum connection and curvature*

$$A^{\infty},\ F^{\infty} \in C^{\infty}(\mathbb{R}^4; \mathfrak{h}_3), \qquad \mathcal{A}^{\infty},\ \mathcal{F}^{\infty} \in C^{\infty}(\mathbb{R}^4; \mathfrak{su}(3)),$$

with the Hermitian and standard anti-Hermitian $SU(3)$ conventions identified explicitly.

(iv) ***Local Wilson-to-Yang–Mills limit.*** *Theorem 7.2 proves that for every $\chi \in C_c(\mathbb{R}^4)$, the localized Wilson functional converges to the localized continuum Yang–Mills functional*

$$S^{(N)}_{\mathrm{W},\chi} \longrightarrow S^{\mathrm{cont}}_{\mathrm{YM},\chi}[F^{\infty}].$$

(v) ***Global cutoff-free limit.*** *Theorem 7.5 proves that the full Wilson action on the deterministic flat regulator converges to the global continuum Yang–Mills functional*

$$S_{\mathrm{W}}^{(N)} \longrightarrow S_{\mathrm{YM}}^{\mathrm{cont}}[F^{\infty}].$$

Accordingly, downstream papers may cite the present paper as establishing the deterministic flat regulator, the sampled $SU(3)$ link/plaquette sector, the smooth continuum curvature profile, and the local and global Wilson-to-Yang–Mills continuum-limit theorems.

Proof. Item (i) is exactly the collection of labeled deterministic regulator constructions Definitions 4.1, 5.1, 5.3, and 5.6. Item (ii) is exactly the sampled lattice gauge package of Definitions 6.11 and 6.14 together with Proposition 6.15. Item (iii) is the smooth continuum gauge package of Definition 3.1, Propositions 6.3, 6.5, and 6.15. Item (iv) is Theorem 7.2, and item (v) is Theorem 7.5. The closing sentence is only a summary of these theorem-level outputs. □

Remark 8.2 (How later papers use this export package)*.* Paper 3 imports the stationary continuum-indexed gauge observables built from the deterministic flat regulator and the corresponding continuum Yang–Mills functional. Paper 5 then imports from the present paper exactly the regulator geometry, sampled $SU(3)$ gauge sector, smooth continuum curvature, and local/global Wilson-to-Yang–Mills limit recorded in Theorem 8.1.

References

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[2] P. Diaconis and D. Freedman, "Finite exchangeable sequences," *The Annals of Probability* **8** (1980), no. 4, 745–764.